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IEEE Guide for Acceptance and Maintenance of Insulating Mineral Oil in Electrical Equipment

IEEE Power and Energy Society

Sponsored by the
Transformers Committee

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USA

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(Revision of
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Abstract: Recommendations regarding mineral oil tests and evaluation procedures are made in this guide; references are made to methods of reconditioning and reclaiming conventional petroleum (mineral) dielectric insulating liquids; the levels at which these methods become necessary; and the routines for restoring oxidation resistance, where required, by the addition of oxidation inhibitors. The intent is to assist the power equipment operator in evaluating the serviceability of mineral oil received in equipment, oil as received from the supplier for filling new equipment at the installation site, and oil as processed into such equipment; and to assist the operator in maintaining mineral oil in serviceable condition. The mineral oil covered is used in transformers, reactors, load tap changers, and voltage regulators.

Keywords: IEEE C57.106™, insulation testing, load tap changers, mineral oil insulation, power distribution maintenance, power transformer insulation, reactors, transformers, voltage regulators

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Introduction

This introduction is not part of IEEE Std C57.106™-2015, IEEE Guide for Acceptance and Maintenance of Insulating Mineral Oil in Electrical Equipment.

IEEE Std C57.106™ was last revised in 2006. There were several noteworthy changes to this guide in the 2015 revision. The overall outline common to many guides, including this one, was not altered; but subclauses were arranged to improve the flow of information. An example is the movement of the in-service oil subclause to just after the new oil subclause.

In this revision, discussions addressing moisture in the solid insulation/paper insulation were removed. In addition, information on oil circuit breakers was removed and accepted by the switchgear C37.20 series.

In general, terms were aligned to recommended nomenclature. The guide was made responsive to consistent terms for the oils, e.g., using the name "mineral oil" as opposed to several others. When necessary or practical, the references to a specific agency, title, era, or geographic specific rule were substituted with wording such as "the local, state, and national regulations."

Contents

IEEE Guide for Acceptance and Maintenance of Insulating Mineral Oil in Electrical Equipment

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1. Overview

The reliable performance of oil in insulation systems depends on the basic characteristics of the oil that can affect overall apparatus characteristics. These oil characteristics are integral parts of the equipment design of the manufacturer. Certain properties of mineral insulating oil have been determined as important for proper electrical equipment performance. A description of these properties and their recommended value ranges for new oil and for continued use of in-service oils are included in this guide.

Mineral insulating oil that is received in electrical equipment could exhibit different characteristics from new oil received in bulk, which has not been in contact with apparatus construction materials.

Oil in service may contain dissolved gases that are useful in assessing the continued serviceability of certain types of transformers. It is not the intent of this guide to cover this subject, as that information is available in IEEE Std C57.104™.

Should instructions or product standards given by the manufacturer differ from recommendations made in this guide, the instructions of the manufacturer are to be given preference.

1.1 Scope

This guide applies to mineral oil used in transformers, load tap changers, voltage regulators, and reactors. The guide discusses the following:

a) Analytical tests and their significance for the evaluation of mineral insulating oil.

b) The evaluation of new, unused mineral insulating oil before and after filling into equipment.

c) The evaluation of in-service mineral insulating oil.

d) Health and environmental care procedures for mineral insulating oil.

e) Methods of handling and storage of mineral insulating oil.

The characteristics of the mineral oils discussed in this guide do not include reclaimed or reconditioned oil installed in new equipment. The qualities of such oil, if used, should be agreed upon by the manufacturer and the user of the equipment.

1.2 Purpose

The purpose of this guide is to assist the user of the equipment in evaluating the serviceability of new, unused mineral oil being received in equipment; mineral oil as received for filling new equipment at the installation site; mineral oil as processed into equipment; and in-service mineral oil in equipment. It also assists the operator in maintaining the mineral oil in serviceable condition.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

ASTM D3487, Standard Specification for Mineral Insulating Oil Used in Electrical Apparatus.

IEEE Std 637™, IEEE Guide for the Reclamation of Insulating Oil and Criteria for its Use.[1,2]

IEEE Std C57.93™, IEEE Guide for Installation of Liquid-Immersed Power Transformers.

IEEE Std C57.131™, IEEE Standard Requirements for Load Tap Changers.

IEEE Std C57.12.90™, IEEE Standard Test Code for Liquid-Immersed Distribution, Power, and Regulating Transformers.

3. Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause. [3]

electrical equipment: For purposes of this guide, electrical equipment refers to transformers, load tap changers (LTCs), voltage regulators, and reactors.

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[3] *IEEE Standards Dictionary Online* subscription is available at: http://www.ieee.org/portal/innovate/products/standard/standards_dictionary.html.

mineral oil: A specially refined oil of petroleum origin for use as an insulating liquid and coolant in transformers and other electrical equipment. Generally conforms to ASTM D3487 when new.

NOTE—This guide uses several different terms to identify insulating oil of a petroleum origin. The terms oil, insulating oil, liquid, mineral insulating oil, conventional mineral oil, and dielectric liquid are all equivalent to mineral oil in the context of this guide.[4]

percent relative saturation (%RS): The ratio, expressed as a percentage, between the actual moisture in oil concentration and the saturation value (also called the solubility limit) of moisture in oil at a given temperature.

reclamation of mineral oil: The restoration to usefulness by the removal of contaminants and products of degradation such as polar, acidic, or colloidal materials from used electrical insulating fluids by chemical or adsorbent means. Reclaiming typically includes treatment with clay or other adsorbents. *See also:* **reconditioning**.

NOTE—The methods listed under reconditioning are usually performed in conjunction with reclaiming.

reconditioning of mineral oil: The removal of insoluble contaminants, moisture, and dissolved gases from used electrical insulating fluids by mechanical means.

NOTE—The typical means employed are settling, filtering, centrifuging, and vacuum drying or degassing.

solubility limit: A quantity expressing the maximum concentration of water that can exist in insulating liquid at specified temperature. *Syn:* **saturation value**.

vacuum filling: The process whereby the liquid-impregnated insulation is filled or re-filled with dielectric liquid under vacuum.

NOTE—This process involves the exposure of the core and coil to vacuum prior to filling for time periods based on either manufacturer standards or IEEE Std C57.93™.

4. Evaluation of mineral insulating oil

4.1 New mineral oil properties

New mineral insulating oils as delivered must conform to the property requirements listed in ASTM D3487.[5] Newly supplied oils have many characteristics related to their chemical and molecular structure that are directly measured by test methods such as viscosity, flash and fire points, pour point, aniline point, relative density (specific gravity), oxidation stability, gassing tendency, and dielectric breakdown voltage.

Many characteristics not necessarily related to the functional performance of mineral insulating oils are evaluated because of their sensitivity to the presence of harmful contaminants. Some of the following characteristics are sensitive to contamination in the oil: interfacial tension, dissipation factor (power factor), dielectric breakdown voltage, color, and neutralization number (acidity).

[4] Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.
[5] Information on references can be found in Clause 2.

4.1.1 New mineral oil properties—as supplied

New mineral insulating oils as supplied must have properties so that new oils meet the standard specification of ASTM D3487 and Table 1 of this guide when received, prior to any processing. When mineral insulating oil specified to conform to ASTM D3487 is received, it should be tested to verify conformance with ASTM D3487. Testing of the oil for full conformance of all property requirements of ASTM D3487 is only justified under circumstances determined by the purchaser. It is recommended that the purchaser require the supplier to provide a certified set of tests for the oil that demonstrate that the oil, as shipped, met or exceeded the property requirements of ASTM D3487. For those circumstances where a full set of tests according to ASTM D3487 are not justified, it is recommended that, at a minimum, the tests shown in Table 1 of this guide be considered. The purchaser of the oil should conduct tests sufficient to satisfy concerns regarding conditions of shipment that might result in nonconformance to ASTM D3487 property requirements. Table 1 lists several of the more important tests with values that should help in the decision regarding acceptance of the new mineral insulating oil.

Insulating liquid is ordinarily shipped in four types of containers: drums, totes, tank trailers, and rail cars. Rail cars are usually under the control of the supplier and dedicated to insulating oil shipment. Highway trailers are used to transport many different chemical products as well as insulating oil; these trailers are therefore subject to chemical contamination. Special cleaning and drying procedures may be necessary. If problems are encountered, check the history of the shipping containers to see that they have been cared for properly. It is recommended that the purchaser require the delivery of mineral oil in containers that are properly cleaned to guarantee delivery of oil conforming to ASTM D3487.

For large quantity shipments, bulk shipment is preferred. For smaller shipments, drums and totes are often used. Drums and totes should be stored under cover to prevent contamination by moisture. Before processing, it is necessary to check the quality of the oil in each drum or tote or after blending the oil in a large tank. Each tank load or each shipping unit of oil as received at the customer's site should undergo a check test to determine that the electrical characteristics have not been impaired during transit or storage. Table 1 contains a list of recommended acceptance tests for shipments of mineral insulating oil as received from the supplier. Some users may not wish to perform all these tests; however, at a minimum, dielectric breakdown voltage and dissipation factor (power factor) as listed in Table 1 should be performed. It is satisfactory to accept oils that exhibit characteristics other than those described by the values in Table 1 providing that the users and the suppliers are in agreement.

Table 1—Test limits for shipments of new mineral oil as received from the oil supplier

Test and method	Limit value
Dielectric breakdown voltage ASTM D1816 kV minimum 1 mm gap 2 mm gap	 20 35
Dissipation factor (power factor) ASTM D924 25 °C, % maximum 100 °C, % maximum	 0.05 0.30
Interfacial tension ASTM D971 mN/m minimum	 40
Color ASTM D1500 ASTM units maximum	 0.5
Visual examination ASTM D1524	 Bright and clear
Neutralization number (acidity) ASTM D974 Mg KOH/g maximum	 0.03
Water content ASTM D1533 Mg/kg maximum[a]	 35
Oxidation inhibitor content when specified ASTM D2668 Type I oil, % maximum Type II oil, % maximum	 0.08 0.3
Corrosive sulfur ASTM D1275	 Noncorrosive
Relative density (specific gravity) ASTM D1298 15 °C maximum	 0.91

[a] Equivalent measurement of mg/kg is parts per million (ppm) on a weight-by-weight basis. From a dielectric breakdown voltage point of view, the moisture dissolved in oil limits given could be excessive at low temperatures due to higher levels of percent saturation. See 4.5.

4.1.2 New mineral oil properties—oil received in new equipment rated less than 230 kV

Mineral oil that has been shipped in new equipment from the manufacturing plant may be evaluated by obtaining a sample from the equipment at the job site. Recommended test limits for mineral oil received in new equipment prior to energization are given in Table 2. Properties, such as interfacial tension (IFT), dielectric breakdown voltage, and dissipation factor, which are sensitive to certain dissolved or particulate contaminants, will reflect the exposure to normal equipment construction materials.

Table 2—Test limits for new mineral oil received in new equipment or after filling, prior to energization

Test and method	Value for voltage class		
	≤69 kV	>69 – <230 kV	≥230 kV
Dielectric breakdown voltage ASTM D1816 kV minimum 1 mm gap 2 mm gap	25 45	30 55	35 60
Dissipation factor (power factor) ASTM D924 25 ºC, % maximum 100 °C, % maximum	0.05 0.40	0.05 0.40	0.05 0.30
Interfacial tension ASTM D971 mN/m minimum	38	38	38
Color ASTM D1500 ASTM units maximum	1.0	1.0	0.5
Visual examination ASTM D1524	Bright and clear	Bright and clear	Bright and clear
Neutralization number (acidity) ASTM D974 mg KOH/g maximum	0.03	0.03	0.03
Water content ASTM D1533 mg/kg maximum[a]	20	10	10
Oxidation inhibitor content when specified ASTM D2668 Type I mineral oil, % maximum Type II mineral oil, % maximum	0.08 0.3	0.08 0.3	0.08 0.3
Total dissolved gas ASTM D3612	N/A	N/A	0.5% or per manufacturer's requirements[b,c]
Corrosive sulfur ASTM D1275	Noncorrosive	Noncorrosive	Noncorrosive

[a] Equivalent measurement of mg/kg is parts per million (ppm) on a weight-by-weight basis. From a dielectric breakdown voltage point of view, the moisture dissolved in mineral oil limits given could be excessive at low temperatures due to higher levels of percent saturation. See 4.5.

[b] This value should be obtained from a sample collected 24 h to 48 h after the transformer is filled and applies only to transformers with diaphragm conservator systems.

[c] One-half percent (0.5%) is used for total dissolved gases, which is the sum of the ppms of the individual gasses which include any atmospheric gasses. 0.5% = 5000 ppm

4.1.3 New mineral oil properties—after filling into new equipment rated at 230 kV or higher

Prior to filling equipment rated 230 kV and above with mineral oil at the installation site, manufacturers generally require rigorous processing of the oil to remove moisture, particulate matter, and gas that dissolved during shipment. The object is to fill high-voltage equipment with oil that contains the very least amount of particulate (see also IEEE Std C57.152™) material and water, recognizing that a slight reduction in quality due to contact with the equipment insulation and other materials will occur. In this procedure, oil samples are taken after the oil enters into the electrical equipment.

When mineral oil is received in bulk shipping containers, it is processed according to the instructions of the manufacturer and then introduced into the equipment. Table 2 contains test limits that ensure that the insulating oil, in equipment after the processing and standing time before energization, is dry, contains no excess particulate matter, and contains a minimum amount of dissolved gas.

4.2 In-service mineral oil properties

This subject is treated in greater detail in IEEE Std C57.637™. It is not practical to indicate the value of specific tests and recommended test limits for all possible existing applications of in-service insulating liquid. It should be recognized that, with the current state of knowledge, no single test can be used as the sole criterion to estimate the condition of in-service mineral oil. However, it is possible to summarize the value and importance of the current tests and to suggest methods of treatment for the mineral oil being examined. Any significant changes from previous test data should be investigated. In-service mineral oil may be placed in the classifications described in 4.2.1 based on the composite evaluation of significant characteristics.

Transformer mineral oil is subject to oxidation. Oxidation occurs when sufficient levels of oxygen, moisture, and heat are present. Not all three of these factors are easy to eliminate in a liquid-immersed transformer. For these reasons, providers offer mineral oil with oxidation inhibitor at the levels shown in Table 1 and Table 2.

4.2.1 In-service mineral oil classification

4.2.1.1 Class I

This group contains mineral oils that are in satisfactory condition for continued use as shown in Table 3. It is not intended that oil be removed from service when a single property limit is exceeded or that the oil be left in service until all property values are outside the stated limits.

4.2.1.2 Class II

This group contains mineral oils that do not meet the dielectric breakdown voltage and/or water content requirements of Table 3 and should be reconditioned by mechanical filtration (filter pressing, vacuum dehydration, or similar filtering equipment). For further information on reconditioning, see 4.2 and 5.2 of IEEE Std C57.637.

4.2.1.3 Class III

This group contains mineral oils in poor condition that do not meet the limits in Table 3. This class oil should be reclaimed using Fuller's earth or an equivalent method, or replaced per 4.2 of IEEE Std C57.637. Oil reclamation can remove existing oxidation inhibitor which should be replaced as close to the maximum value as possible (see Table 2).

For information and advice concerning the need and different techniques of reconditioning and reclaiming in-service mineral oils, refer to IEEE Std C57.637. Reconditioning or reclamation of mineral oils containing PCBs can be a violation of environmental regulations. The oil PCB concentration should be quantified prior to maintenance to meet regulatory requirements (see 5.3.2).

4.2.2 In-service mineral oil—suggested limits for continued use

Suggested test limits by voltage class for Class I mineral oils in electrical equipment to remain in continued service are given in Table 3. It is difficult to quantify the risk of electrical equipment failure while in service with particular test values. The limits from Table 3 are intended to provide reference points for continued evaluation and testing. Each case should be examined individually, and the advice of the manufacturer may be considered during the evaluation process. In addition to this information, other guides (and electrical test data) can be consulted, such as IEEE Std C57.104, IEEE Std C57.143™, and IEEE Std C57.152.

Table 3—Suggested limits for continued use of in-service mineral oil

Test and method	Value for voltage class		
	≤69 kV	>69 – <230 kV	≥230 kV
Dielectric breakdown voltage ASTM D1816 kV minimum 1 mm gap 2 mm gap	 23 40	 28 47	 30 50
Dielectric breakdown voltage ASTM D877 (note—not for transformers) 2.54 mm gap [a]	25	25	25
Dissipation factor (power factor) ASTM D924 25 º C, % maximum 100 º C, % maximum	 0.5 5.0	 0.5 5.0	 0.5 5.0
Interfacial tension ASTM D971 mN/m minimum	25	30	32
Neutralization number (acidity) ASTM D974 mg KOH/g maximum	0.20	0.15	0.10
Water content ASTM D1533 mg/kg maximum (ppm) [b]	35	25	20
Oxidation inhibitor content ASTM D2668 Type II mineral oil	0.08% minimum if in original oil		

[a] This applies only to reactors and voltage regulators, not to transformers.

[b] Equivalent measurement of mg/kg is parts per million (ppm) on a weight-by-weight basis. It is important to consider from a dielectric breakdown voltage point of view that the moisture dissolved in mineral oil limits given could be excessive at low temperatures due to higher levels of percent saturation. See 4.5.

4.2.3 Reconditioned or reclaimed mineral oil—suggested limits after filling but before energizing

The suggested test limits in Table 4 may be used after reconditioning or reclaiming in-service mineral oil in transformers or reactors after filling but before energizing. Table 4 is not intended for new equipment. For more information on the reconditioning or reclaiming procedures, the test methods used to evaluate the progress and end point of the process, and what criteria recommended for reclaimed oils are considered suitable, see IEEE Std C57.637. In addition, some transformers are equipped with diaphragms to prevent the introduction of air. The dissolved gas content in these transformers should be maintained in accordance with the manufacturer's recommended limits. Methods and procedures for filling are covered in IEEE Std C57.93.

Table 4 —Suggested test limits for reconditioned or reclaimed mineral oil in transformers and reactors after filling but before energizing

Test and method	Value for voltage class		
	≤69 kV	>69 – <230 kV	≥230 kV
Dielectric breakdown voltage ASTM D1816 kV minimum 1 mm gap 2 mm gap	 25 45	 30 55	 35 60
Interfacial tension ASTM D971 mN/m minimum	35	35	35
Color ASTM D1500 maximum[a]	1.5	1.5	1.5
Visual examination	Clear	Clear	Clear
Neutralization number (acidity) ASTM D974 mg KOH/g maximum	0.05	0.05	0.05
Water content ASTM D1533 mg/kg maximum (ppm)[b]	35	20	15 (<10 desirable)

[a] The color may not always improve to these levels even though the other oil test results may be acceptable.

[b] Equivalent measurement of mg/kg is parts per million (ppm) on a weight-by-weight basis. It is important to consider from a dielectric breakdown voltage point of view that the moisture dissolved in mineral oil limits given could be excessive at low temperatures due to higher levels of percent saturation. See 4.5.

4.3 Mixtures of mineral oil

Although conventional mineral oils from different suppliers may differ in their characteristics, these oils, as well as blends of these oils, should all meet the specifications of ASTM D3487.

Mineral oil is miscible with some types of less-flammable and nonflammable insulating liquids. It is advisable to use dedicated processing and handling systems for each type of liquid to avoid even traces of cross-contamination between petroleum-based and nonpetroleum-based liquids. Traces of silicone liquids can lead to foaming in the oil during vacuum dehydration and filling of equipment. Mixtures of less-flammable (see IEEE Std C57.121™) or nonflammable insulating liquids with conventional mineral oil will alter the flammability characteristics of the less-flammable liquid. Differences in the dielectric constant of the liquids may cause localized voltage stresses in equipment.

CAUTION

Mineral oil is not miscible with silicone insulating liquid. Trace amounts of silicone in mineral oil can cause excessive foaming during degassing process. Always refer to transformer manufacturer recommendation before using or adding an insulating liquid type that is different from the original type.

4.4 Sampling

Accurate sampling methods are extremely important due to their effect on the reliability of the test results for the sampled mineral oil. Careless sampling procedures will result in contamination of the sample, which will lead to erroneous conclusions concerning the quality of the oil. To minimize the possibility of obtaining a non-representative sample, the procedures and precautions outlined in the latest revision of ASTM D923 should be followed.

WARNING

Caution should be taken when extracting an oil sample from energized equipment. Taking samples from equipment under negative pressure (vacuum) may cause an air bubble to be drawn into the tank. If this occurs, it can cause catastrophic failure and death.

4.5 Relative saturation of water dissolved in mineral oil

With regard to dissolved water in mineral oil, solubility equations have been used to assess the saturation limit of water in mineral oil. The saturation limit is the maximum amount of moisture that is soluble in the oil at a specific temperature. Water in oil concentrations in excess of this limit will form free water (water droplets). The ratio of the actual water concentration to its saturation limit is called percent relative saturation (%RS). The equation is [(mg/kg water in oil)/(mg/kg of water in oil at saturation)] × 100.

WARNING

Once the moisture in oil reaches the 100% relative saturation, free water (water droplets) will form, which adversely affects the dielectric breakdown voltage of the oil and may result in transformer failure.

Equations for calculation of water in oil saturation limit at various temperatures and for various mineral oils are found in [B35] and [B43].[6] The relationship between the water in oil solubility and temperature is logarithmic: $\log(S) = A\text{-}B/T$. These references indicate that the water in oil solubility (Factor A and B) can vary between new mineral oil types as well as between new and in-service mineral oil.

As the saturation limit changes with temperature, a given moisture level (in terms of ppm) will result in different %RS. As temperature increases, the saturation limit increases, resulting in a lower %RS. On the other hand, when temperature decreases, the %RS increases.

It is this last phenomenon that may be of concern for a transformer. Lower mineral oil temperatures in a de-energized transformer could increase the %RS level enough to reduce the dielectric breakdown strength of the oil, even at percent saturation levels below 100%. Consequently, there is a concern when energizing transformers at lower ambient temperatures. Caution should be taken so that the dielectric breakdown voltage of the oil, which is referenced to the actual temperature of the oil in the de-energized transformer, is sufficient for service.

Annex B provides an example of the use of published papers to help address these risks [B1], [B36], [B42]. For more information on suggested procedures for energization of mineral oil transformers under cold conditions, see IEEE Std C57.93.

5. Mineral oil tests and their significance

Many established ASTM tests of practical significance can be applied to mineral oil. Some of these tests are more applicable to new oils than to service-aged oils; some are more useful in the analysis of service-aged oils than oils received in new condition, particularly trace contamination between petroleum-based and non-petroleum-based liquids.

Each of the tests and its significance are described in the subclauses of this guide as listed in Table 5. The number after the name of the test is the number of the ASTM standard/method.

Table 5 —Pertinent ASTM tests for mineral oil

5.1 Physical tests	**Test**	**ASTM standard/method**
5.1.1	Aniline point	ASTM D611
5.1.2	Color	ASTM D1500
5.1.3	Flash and fire points	ASTM D92
5.1.4	Interfacial tension	ASTM D971
5.1.5	Pour point	ASTM D97
5.1.6	Relative density (specific gravity)	ASTM D1298
5.1.7	Viscosity	ASTM D445 ASTM D2161
5.1.8	Visual examination	ASTM D1524
5.2 Electrical tests		

[6] The numbers in brackets correspond to those of the bibliography in Annex A.

Table 5—Pertinent ASTM tests for mineral oil, *continued*

	Test	ASTM standard/method
5.2.1	Dielectric breakdown voltage	ASTM D877 ASTM D1816
5.2.2	Dielectric breakdown impulse voltage	ASTM D3300
5.2.3	Dissipation factor (power factor)	ASTM D924
5.2.4	Gassing of electrical insulating liquids under electrical stress and ionization	ASTM D2300
5.3 Chemical tests		
5.3.1	Gas content	ASTM D2945 ASTM D3284 ASTM D3612
5.3.2	Polychlorinated biphenyls	ASTM D4059
5.3.3	Corrosive sulfur	ASTM D1275
5.3.4	Neutralization number (acidity)	ASTM D664 ASTM D974
5.3.5	Oxidation inhibitor content	ASTM D2668 ASTM D4768
5.3.6	Oxidation stability (inhibited oil only, pressure vessel) Oxidation stability	ASTM D2112 ASTM D2440
5.3.7	Water in insulating liquids	ASTM D1533
5.3.8	Furans in insulating liquids	ASTM D5837
5.3.9	Gassing of insulating liquids under thermal stress	ASTM D7150

5.1 Physical tests

5.1.1 Aniline point—ASTM D611

The aniline point (temperature) of mineral oil indicates the solvency of the oil for some materials that are in contact with the oil. A high aniline point indicates a lower degree of aromaticity and a lower solvency for some material (e.g., rubber).

5.1.2 Color—ASTM D1500

Mineral oil should have a light color and be optically clear so that it permits visual inspection of the assembled apparatus inside the equipment tank. Any change in the color of oil over time is an indication of oxidation, deterioration, or contamination of the oil.

5.1.3 Flash and fire points—ASTM D92

The flash point of mineral oil is the temperature to which it must be heated (under prescribed conditions of test) in order to give off sufficient vapor to form a flammable mixture with air. The fire point is the temperature that provides sufficient combustible vapors to sustain a fire for 5 s (under the same test conditions). A low flash point may indicate the presence of volatile combustible contaminants in the mineral oil.

5.1.4 Interfacial tension—ASTM D971

This method covers the measurement, under non-equilibrium conditions, of the surface tension that an insulating liquid maintains against water. Interfacial tension is a measurement of the forces of attraction between molecules of the two liquids. It is expressed in milli-Newton per meter (mN/m). The test is an excellent means of detecting oil-soluble polar contaminants and oxidation products in mineral oils.

5.1.5 Pour point—ASTM D97

The pour point is the temperature at which insulating liquid ceases to flow under prescribed testing conditions. The pour point has little significance as a test for contamination or deterioration of the insulating liquid. It may be useful for oil identification and determination of suitability for a particular climate.

5.1.6 Relative density (specific gravity)—ASTM D1298

The relative density of mineral oil is the ratio of the weights of equal volumes of the oil and water, tested at 15 °C. The relative density is significant in determining the suitability for use in certain applications. When considered along with other oil properties, relative density can be an indicator of the quality of the oil. With a very high specific gravity and at very cold temperatures, ice crystals may form within the transformer.

5.1.7 Viscosity—ASTM D445 and ASTM D2161

The viscosity of insulating liquid is measured by timing the flow of a known volume of insulating liquid through a calibrated tube at a specific temperature. Viscosity has an important influence on the heat transfer characteristics of oil. High viscosity decreases the cooling efficiency of the oil. High viscosity will also affect the movement of parts in electrical equipment, such as circuit breakers, switchgear, tap changers, pumps, and regulators. Viscosity is a factor in determining the conditions for oil processing and cellulose impregnation time.

5.1.8 Visual examination—ASTM D1524

This test indicates the color and degree of turbidity of mineral oil, which may indicate the presence of free water or contaminating solid particles. The source of insoluble solid contaminants may be determined by filtering the particles and examining them. This test may be used to suggest the need for additional laboratory tests as it may permit a determination of whether the sample should be sent to a central laboratory for a full evaluation.

5.2 Electrical tests

5.2.1 Dielectric breakdown voltage—ASTM D877 and ASTM D1816

The dielectric breakdown voltage of insulating liquid is a measure of its ability to withstand voltage stress without failure. It is the voltage at which breakdown occurs between two electrodes under prescribed test conditions. The test serves primarily to indicate the presence of electrically conductive contaminants in the liquid, such as water, dirt, moist cellulosic fibers, or particulate matter. However, a high dielectric breakdown voltage does not indicate the absence of all contaminants.

Two methods are recognized for measuring the dielectric breakdown voltage of insulating liquids as follows:

a) ASTM D877 uses thin flat-faced cylindrical electrodes with a 2.5 mm gap. The sensitivity of this method, to the general population of contaminants present in a liquid sample, decreases as applied test voltages used in this method become greater than 25 kV rms.

b) ASTM D1816 uses spherically-shaped electrodes. The liquid sample is circulated continuously in the test cell throughout the test. The gap distance standard settings are 1 mm and 2 mm.

The rounded electrodes (VDE) used in ASTM D1816 generate a more representative electrical field, closer to those present in a transformer. ASTM D1816 is more responsive to particles and dissolved water, both of which are detrimental to the transformer's oil electrical strength. Therefore, test results from ASTM D1816 furnish a better evaluation of changes that may occur in transformer oil. The flat disk electrodes used in ASTM D877 generate an electrical field that is not representative of those normally present in a transformer.

5.2.2 Dielectric breakdown impulse voltage—ASTM D3300

This test method is most commonly performed using a negative polarity point opposing a grounded sphere (NPS). The NPS breakdown voltage of fresh unused mineral oils measured in the highly divergent field in this configuration depends on oil composition; decreasing with increasing concentration of aromatic molecules, and particularly polyaromatic molecules.

This test method may be used to evaluate the continuity of composition of oil from shipment to shipment. The NPS impulse breakdown voltage of oil can also be substantially lowered by contact with materials of construction, service aging, and other impurities. Test results lower than those expected for fresh oil may also indicate use or contamination of that oil.

Although polarity of the voltage wave has little or no effect on the breakdown strength of oil in uniform fields, polarity does have a marked effect on the breakdown voltage of oil in non-uniform electric fields.

Transient voltages may also vary over a wide range in both the time to reach crest value and the time to decay to half crest or to zero magnitude. The standard lightning impulse test specifies a 1.2×50 μs negative polarity wave as described in IEEE Std C57.12.90™.

5.2.3 Dissipation factor (power factor)—ASTM D924

The dissipation factor is a measure of the power lost when an electrical insulating liquid is subjected to an ac field. The power is dissipated as heat within the liquid. A low-value dissipation factor means that the liquid will cause little of the applied power to be lost. The test is used as a check on the deterioration and contamination of insulating oil because of its sensitivity to ionic contaminants.

5.2.4 Gassing of insulating liquids under electrical stress and ionization—ASTM D2300

This test measures whether insulating liquids are gas absorbing or gas evolving when subjected to electrical voltage. For certain applications, when insulating liquids are stressed at high voltage gradients, it is desirable to know the rate at which gas is absorbed or evolved from the liquid. The absorption or evolution of gas by a liquid under electrical stress is a function of the aromatic character of the liquid molecules. Liquids that are significantly aromatic in character will absorb gas as they are electrically stressed. Liquids that have little or no aromatic character will evolve hydrogen gas upon application of an electrical voltage. Currently, however, correlation of these test results with equipment performance is limited. Numerical results obtained in different laboratories or by using two different procedures may differ significantly in magnitude, and the results of this method should be considered qualitative in nature.

5.3 Chemical tests

5.3.1 Gas content—ASTM D3612

The gas content of an insulating liquid may be defined as the percentage volume of dissolved gas or by ppm (parts per million) at standard pressure and temperature. Some types of equipment require the use of electrical insulating liquids of low gas content. In filling electrical apparatus, low gas content reduces foaming and reduces the available oxygen, thereby increasing the service life of the insulating liquid.

Although not initially intended for this sole purpose, ASTM D3612 [B29] could be used to measure the total gas content in mineral oil as an alternate to ASTM D2945 [B25] that has been withdrawn. Gas content is not intended for use in purchase specifications because transformer oil is customarily degassed immediately prior to use.

ASTM D3612 can also be used as a factory control test and is useful in evaluating the health of the transformer equipment. Overheating, partial discharges, or arcing within the transformer will generate combustible and noncombustible gasses that will be dissolved in the mineral oil and could be measured by ASTM D3612.

The amount and kinds of gases dissolved in mineral oil can be used as a tool to aid in detecting and diagnosing faults and abnormal operating conditions in equipment. For dissolved gas analysis interpretation, refer to IEEE Std C57.104 [B39] and IEEE Std C57.139 [B41].

5.3.2 Polychlorinated biphenyls (PCBs)—ASTM D4059

Throughout the world, regulations require that electrical apparatus and electrical insulating liquids containing polychlorinated biphenyls (PCBs) be handled and disposed of through the use of specific procedures. The procedure to be used for a particular apparatus or quantity of insulating liquid is determined by the PCB content of the liquid. The results of this analytical technique can be useful in selecting the appropriate handling and disposal procedures. Handling and disposing of electrical apparatus and electrical insulating liquids containing polychlorinated biphenyls is regulated in most countries by the national, state, or local regulation. For instance, in the U.S., refer to Title 40 CFR, Part 761 [B47].

5.3.3 Corrosive sulfur—ASTM D1275

This test is used to determine the presence of corrosive sulfur in mineral oil. The mineral oil is heated under established conditions with a polished copper coupon (and in one test with additional paper). The copper coupon (and paper) is inspected for the characteristic appearances of corrosive sulfur [B33]. One of the sources of sulfur which can lead to corrosive sulfur present in mineral oil can be the crude oil from which it was refined if it was not severely hydro-treated. The sulfur may also come from other sources, such as rubber hoses used for oil processing or from replacement gasket material. In addition, one other important source has been identified. Dibenzyldisulfide (DBDS), which was used for a period as an oxidation inhibitor enhancer, has been found to produce corrosive sulfur in operating transformers.

5.3.4 Neutralization number (acidity)—ASTM D664 and ASTM D974

The neutralization number of an electrical insulating liquid is a measure of the acidic components of that material. In new oil, any acid present is likely residual from the petroleum refining process. In a service-aged liquid, the neutralization number is a measure of the acidic by-products of the oxidation of oil. The neutralization number may be used as a general guide for determining when oil should be reprocessed or replaced. ASTM D974 [B12] is the traditional color-change indicator method of titrating the acids with a mild (0.1 N) KOH solution. ASTM D664 [B7] is a potentiometric titration method. On some service-aged liquids, the color may be so dark as to impair the ability of the technician to determine the indicator color change in ASTM D974, so ASTM D664 is used instead. The correlation between these two methods, however, has not been established.

5.3.5 Oxidation inhibitor content—ASTM D2668 by infrared spectrophotometry and ASTM D4768 by gas chromatography

Two synthetic oxidation inhibitors are commonly used in dielectric liquids. They are 2-6 ditertiary-butyl phenol (DBP) and 2-6 ditertiary-butyl para-cresol (DBPC). Their use provides added resistance to oxidation in systems that are partially or wholly exposed to air. The effectiveness of the oxidation inhibitor depends a great deal on the type of crude oil from which the insulating mineral oil came. Certain new mineral oils may contain naturally occurring antioxidant substances that may yield a false-positive indication in this test.

5.3.6 Oxidation stability—ASTM D2112 and ASTM D2440

Oxidation stability—ASTM D2112: This test method is a rapid test for evaluating the oxidation stability of a new mineral oil that contains the synthetic oxidation inhibitor 2-6 DBPC or 2-6 DBP. The test measures the length of time required for the oil sample to react with a given volume of oxygen when a sample of oil is heated and oxidized under test conditions.

Oxidation stability—ASTM D2440: This test method determines the resistance of mineral oils to oxidation under prescribed accelerated aging conditions. Oxidation stability is measured by the propensity of oils to form sludge and acid products during oxidation. This test method is applicable to new oils, both inhibited and uninhibited.

5.3.7 Water content in insulating liquid—ASTM D1533

Water in insulating liquid may be present in several forms: dissolved water, water in suspension (emulsion), free water in the form of water droplets or water accumulated at the bottom, and water bound to polar compounds, existing due to insulating liquid aging and deterioration. According to ASTM D1533, water content of the insulating liquid is measured in ppm by weight or mg/kg. However, to estimate the effect of water on dielectric breakdown strength, the %RS value is a more adequate quantity to use as explained in 4.5. Thus, the importance of measuring water content in insulating liquid is in measuring and comparing water content with the values given in Table 1 through Table 4.

5.3.8 Furans in insulating liquids—ASTM D5837

Furanic compounds are generated by the degradation of cellulosic materials used in the solid insulation systems of electrical equipment. Furanic compounds that are oil soluble to an appreciable degree will migrate into the insulating liquid. The presence of high concentrations of furanic compounds is significant in that this may be an indication of cellulose degradation from aging or incipient fault conditions. Testing for furanic compounds by high-performance liquid chromatography (HPLC) may be used to complement dissolved gas in oil analysis as performed in accordance with the test method in ASTM D3612. Solvent refined mineral oil could contain traces of furans [B37].

5.3.9 Determination of gassing characteristic of insulating liquids under thermal stress—ASTM D7150

Some insulating liquids subjected to thermal stress, liquids that contain certain types of contamination, and some new liquids may produce specific gases, especially hydrogen, at lower temperatures than normally expected for their generation. This may falsely indicate abnormal operation of the electrical apparatus. Recently, this phenomenon, commonly called “stray gassing,” has been observed more frequently. This renders interpretation of the dissolved gas analysis more complicated.

This test method determines the gassing characteristics due to thermal stress at 120 °C of insulating liquids specifically and without the influence of other transformer materials or electrical stresses.

6. Handling and storage

It is not intended that the recommendations given in this guide supersede any federal, state, or local regulations concerning storage, handling, or spill cleanup of insulating liquids. For further information, refer to IEEE Std 980™.

6.1 Tanks

For new installations, direct transfer of the mineral oil from on-site delivery containers into equipment, although recommended, is not always practical if the oil being delivered is to be tested before transfer into the equipment. Repairs also often include draining mineral oil from the equipment into a suitable tank or tanker for temporary storage.

6.1.1 Use of storage tanks

In these instances, all tanks should conform to applicable national and local standards and codes. Tanks should be equipped with at least one manhole. The interior of the tanks should be stainless steel or sandblasted, primed, and coated with a coating that is compatible with insulating liquids. Storage tanks should be equipped with bottom drains to allow complete emptying of the tank. Vents should be equipped with desiccant dryer, or the tank should have a dry gas blanket.

6.1.2 Use of mobile tankers

There are two major kinds of tankers commonly used with insulating liquids. These have the same function as site storage tanks but are obviously mobile.

a) Most mineral oil is delivered in mobile tandem axle trailers holding from 19 000 to 22 500 liters (5000 to 6000 gallons). The tankers are equipped with varying valve configurations, but normally a connection at the bottom of the tanker is larger than the 51 mm (2 in) or smaller hoses that will transport the oil to the vacuum oil purifier used for most oil handling and filling procedures.

b) A frac tanker is a larger, usually double-walled, 64 000 liter (17 000 gallon) tank that is delivered to the site empty. It is not transferred full of oil. It is useful because of its larger size and more economical demurrage rates while in use. It is more common to see these containers used in repair, since oil is not delivered in such portable tankers. The 80 000 liter (21 000 gallon) type is not double-walled. Tankers, like storage containers, must be thoroughly cleaned, wiped dry with clean rags, and flushed with clean insulating oil. A certificate of cleanliness should be a part of the contractual requirements if the tanker is rented.

6.1.3 Use of storage bladders

Collapsible or rubber fabric tanks may be used for short-time storage. Care should be taken to ensure they are protected from damage and that they are cleaned and thoroughly drained before filling with mineral oil.

a) Position storage bladders where they can be conveniently used as they cannot be moved when oil-filled.

b) Make sure that bladders will not be punctured or damaged (some are tougher than tanks, and some are not).

c) Make sure that bladders cannot roll if on a slightly inclined surface.

6.2 Mineral oil quality protection in storage

Before use, storage tanks should be thoroughly cleaned, wiped dry with clean rags, and flushed with clean mineral oil. It is recommended that insulating oil not be held for more than three months in temporary storage tanks. Where that is unavoidable, tanks should be equipped with a dry inert (nitrogen) gas system with which to blanket the oil.

All storage tanks should be equipped with a dry nitrogen gas supply or a desiccant vent dryer to minimize the introduction of moisture into the tank. Proper maintenance of the desiccant is essential.

6.3 Dikes and curbs

Tanks for storage of insulating liquid may be surrounded by dikes or curbs sufficient to contain the entire volume of liquid in the tank should a spill occur. In the U.S., refer to Title 40 CFR or other applicable national, state, or local requirements for spill prevention, control, and countermeasure plans (SPCC plans).

6.4 Processing mineral oil for installation in apparatus

See IEEE Std C57.93.

Processing systems consist of mineral oil dehydration, degasifying, and filtration. The user should be advised that static charges can develop when insulating oil flows in pipes, hoses, and tanks. Oil leaving a cartridge type filter (in the past often referred to as a filter press) may be charged to a high potential. To accelerate dissipation of the charge in the oil, ground and bond together the filter, the hoses or piping, the equipment tank, and all bushings or the winding. The hoses should be checked at least annually for continuity of the static ground wire or mesh. Conduction through oil is slow; therefore, it is desirable to maintain these grounds for at least an hour after the oil flow has been stopped. Remove any explosive gas mixture from any container into which oil is flowing. Arcs can occur from the free surface of the oil even though the previous grounding precautions have been taken.

Equipment used for handling mineral oils should be dedicated for that use as these oils are very sensitive to contamination. Although mineral oils may be miscible with other types of insulating liquids, it is advisable to use dedicated processing and handling systems for each different type of liquid. When this is unavoidable, there are a few liquids that are soluble with each other to where, in some cases, the processor can be thoroughly flushed and continue in mineral oil service. Silicone insulating liquid should be processed only in dedicated equipment.

Before filling the electrical apparatus, a vacuum may be drawn on its tank. All mineral oil transfer lines should be flushed through with clean, processed oil. During the filling operations, flow rates should be controlled to the value specified by the manufacturer. The vacuum held on the tank should not exceed the design strength specified by the manufacturer. When this information is not available, consult IEEE Std C57.93, where a filling rate is suggested, currently at no more than 0.5 in (13 mm) per min.

The filling process may be varied to suit the capabilities of the equipment used and the recommendations of the manufacturer. Filling procedures used should comply with those that are recommended by the liquid and the equipment manufacturers in order to maintain eligibility of the equipment warranty.

7. Mineral oil for load tap changers (LTCs)

7.1 General

The requirements of mineral oils used in load tap changers (LTCs) are comparable with those of mineral oils used in power transformers. For LTCs where mineral oil is used for arc quenching, the arcing causes contact erosion and oil carbonization which degrade and contaminate the oil. The degree of contamination depends on the operating current and step-voltage of the LTC, the number of operations, and to some degree, on the quality of the insulating oil. LTCs equipped with vacuum interrupters do not generate arcing in oil; therefore, their oil should not be affected in the same manner. For free-breathing LTCs, the desiccant should be checked periodically to prevent moisture ingress.

Maintenance and internal inspection intervals depend on the type of LTC, the LTC rated through-current, the field experience, and the individual operating conditions. They are suggested as periodic measures with respect to a certain number of operations or after a certain operating time, whichever comes first. The recommended maintenance intervals for an individual LTC type are given in the operating and inspection manuals available for each LTC type. In general, at the time of internal inspections it is common practice to change or recondition the oil. See IEEE Std C57.131.

7.2 Testing methods

Testing methods for LTC mineral oils are the same as those used for mineral oils of power transformers. The procedures for LTC sampling and the sampling intervals may vary compared to insulating oils used in power transformers. Refer to ASTM D923 and the LTC operation manual.

7.2.1 New mineral oil properties—as supplied

New mineral oils delivered shall conform to the property requirements listed in ASTM D3487.

7.2.2 Test limits—shipment of new mineral oil

When mineral insulating oil meeting the qualifications of ASTM D3487 is received in the field for installation in LTCs, it should be checked in order for certain key values that may be affected by shipment and storage (Table 1). In cold climates (below -25 °C and above -40 °C) additional values, such as viscosity [12 cSt (12 mm^2/s) maximum at 40 °C] and pour point (–40 °C maximum) should be checked to ensure that the oil does not interfere with the free operation of the equipment.

NOTE—It is common practice to specify a lower or higher pour point depending upon climatic conditions.

7.2.3 New mineral oil properties—oil shipped in new equipment

LTCs are normally shipped from the factory without mineral oil. If an LTC is filled with mineral oil prior to shipping, refer to Table 6.

7.2.4 New mineral oil properties—prior to energizing

When mineral oil is received in bulk shipping containers, it is processed according to the instructions of the LTC manufacturer, then introduced into the equipment. Table 6 gives test limits for LTC mineral oil after being processed, placed in equipment, and allowing for standing time before energizing.

Table 6 —Test limits of new mineral oil for load tap changers, after processing, prior to energizing

Test and method	Suggested limit	
	≤69 kV	>69 kV
Dielectric breakdown voltage ASTM D1816 kV minimum 1 mm, gap 2 mm, gap	 25 45	 30 55
Dielectric breakdown voltage ASTM D877 kV minimum[a]	30	30
Neutralization number (acidity) ASTM D974 mg KOH/g maximum	0.03	0.03
Water content ASTM D1533 mg/kg maximum[b]	20	10
Corrosive sulfur[c] ASTM D1275	Noncorrosive	Noncorrosive

[a] Perform the dielectric breakdown voltage test using either ASTM D1816 or ASTM D877 as shown.

[b] Equivalent measurement of mg/kg is parts per million (ppm) on a weight-by-weight basis.

[c] Under the influence of heat, sulfur containing molecules may decompose and react with metal surfaces to form metal sulfide layers. Such layers can be harmful for switching equipment as they may increase the contact resistance and cause silver-sulfide layers to flake. Other test methods such as IEC 62535 or DIN 51353 [B34] may be appropriate to detect possible reactions of corrosive sulfur with silver.

7.2.5 In-service mineral oil properties

Suggested limits for continued use of in-service LTC mineral oils are shown in Table 7. For information regarding dissolved gas analysis, refer to IEEE Std C57.139.

Table 7 —Limits for continued use of in-service oil for load tap changers

Test and method	Suggested limit		
	Neutral end	Line end	
		≤69 kV	>69 kV
Dielectric breakdown voltage ASTM D1816 kV minimum 1 mm gap 2 mm gap	 20 30	 25 35	 30 45
Dielectric breakdown voltage ASTM D877 kV minimum[a]	25		
Neutralization number (acidity) ASTM D974 mg KOH/g maximum	0.20		
Interfacial tension ASTM D971 mN/m minimum	25		
Color ASTM D1500 ASTM units maximum	2.0		
Water content ASTM D1533 mg/kg maximum[b]	40	30	25

[a] Perform the dielectric breakdown voltage test using either ASTM D1816 or ASTM D877 as shown.

[b] Equivalent measurement of mg/kg is parts per million (ppm) on a weight-by-weight basis.

7.3 Reconditioning

If the dielectric breakdown voltage of the mineral oil drops below the values given in Table 7, or the water content exceeds the values given in Table 7, the mineral oil shall be reconditioned or changed.

8. Health and environmental care procedures for mineral oil

8.1 Health issues

Users should obtain a safety data sheet (SDS) or material safety data sheet (MSDS) for each insulating liquid in use. When instructions differ from recommendations made here, the instructions of the manufacturer are to be followed. Although no special risk is involved in the normal handling of insulating liquids addressed in this guide, attention should be focused on the general need for personal hygiene or the practice of washing skin and clothing that may have come in contact with mineral oil. Personnel should avoid contact of the liquid with their eyes. When dielectric liquids have to be disposed of, certain precautions are necessary to comply with local, state, and national requirements. These liquids may be generally classified as special, regulated, or hazardous waste.

Unless a PCB analysis has been performed, it is prudent to assume that the batch of oil contains PCBs and to act accordingly. The absence of PCBs in a volume of oil in or from a piece of equipment can be established only by analysis of that oil.

8.2 Leaks and spills

During equipment inspection or servicing, routine checks should be made of the equipment and surroundings for leaks. Areas to check and repair should include valves, bushings, gauges, tap changers, welds, sample ports, manhole covers, pipefittings, and pressure relief valves. Refer to IEEE Std 980.

The production of PCBs was banned in 1979; however, many older transformers and other pieces of electrical equipment in service are filled with mineral insulating oil that may contain PCBs. Since 1977, various national, state, and local environmental regulations have governed the handling and processing of mineral oils containing PCBs.

8.2.1 Minor spills

Minor spills, such as those occurring in the manufacture or repair of equipment, can be cleaned using absorbent rags or other materials.

8.2.2 Spills on soil

Soil acts as an absorbent and should not be allowed to become saturated with mineral insulating oil. Users should consult the applicable local, state, and national guidelines for spills of mineral oil onto soil and the remedies available. Depending on local regulations, spills to soil may have to be reported to one or more regulatory agencies.

8.2.3 Spills on water

Local, state, and national guidelines must be followed in the event of a spill for reporting purposes. Because mineral oil floats on water, a spill can be contained by using floating booms or dikes. These systems include pumps, skimmers, physical absorbents, and fibers that are fabricated into floating ropes.

For instance, in the United States, the Clean Water Act regulations are found in Title 40 Code of Federal Regulations, Part 110 and Part 112. These regulations impose reporting requirements for petroleum oils that are spilled into navigable water ways. The requirement to report is triggered by discharges of oil that cause a film or sheen upon, or discoloration of, the surface of the water or adjoining shorelines or cause a sludge or emulsion to be deposited beneath the surface of the water or upon adjoining shorelines. The U.S. Coast Guard and the National Response Center must be notified. If the spill is in a substation, the spill must be reported to the U.S. Environmental Protection Agency as well. More information on spills in substations is available in IEEE Std 980.

Annex A

(informative)

Bibliography

Bibliographical references are resources that provide additional or helpful material but do not need to be understood or used to implement this standard. Reference to these resources is made for informational use only.

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[B2] ASTM D88, Standard Test Method for Saybolt Viscosity.[7]

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[B5] ASTM D445, Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity).

[B6] ASTM D611, Standard Test Methods for Aniline Point and Mixed Aniline Point of Petroleum Products and Hydrocarbon Solvents.

[B7] ASTM D664, Standard Test Method for Acid Number of Petroleum Products by Potentiometric Titration.

[B8] ASTM D877, Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids Using Disk Electrodes.

[B9] ASTM D923, Standard Practice for Sampling Electrical Insulating Liquids.

[B10] ASTM D924, Standard Test Method for Dissipation Factor (or Power Factor) and Relative Permittivity (Dielectric Constant) of Electrical Insulating Liquids.

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[B30] ASTM D4059, Standard Test Method for Analysis of Polychlorinated Biphenyls in Insulating Liquids by Gas Chromatography.

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Annex B

(informative)

Examples ASTM D1816 breakdown voltage vs. %RS for new mineral oil

According to Oommen [B43], Equation (B.1) below is given for the relationship between the percent relative saturation of dissolved water in mineral oil (%RS) and the temperature and parts per million (ppm) of dissolved water in oil. Parameters A and B are the experimentally determined coefficients for new oil where A = 7.42 and B = 1670.

$$\text{Relative Saturation of Dissolved Water in Mineral Oil} = ppm\ H_2O\ in\ oil \div 10^{A-\left(\frac{B}{273.16+\deg.C}\right)} \quad \text{(B.1)}$$

In Miners [B42] and Ast [B1], graphs are given for ASTM D1816 (2 mm gap) dielectric breakdown voltage values vs. ppm (not %RS) of dissolved water in oil. These tests were done at room temperature.

In order to view these dielectric breakdown voltage values in these graphs as a function of %RS, Equation (B.1) can be used to convert the ppm in oil values in these graphs to %RS. Then, a least squares nonlinear regression gives Equation (B.2), which uses an average of the parameters of regression from the graphs in [B42] and [B1].

$$\%\ \text{ASTM D1816 (2mm) Voltage Breakdown} = 0.95 \times e^{-\left(\frac{\%RS}{63.15}\right)^2} + 0.05 \quad \text{(B.2)}$$

Equation (B.2) can be used to construct the graph below. Both %RS (the x-axis) and the ASTM D1816 voltage breakdown values (the y-axis) are normalized to 100% and can be used as a means to graphically illustrate the values presented in the references.

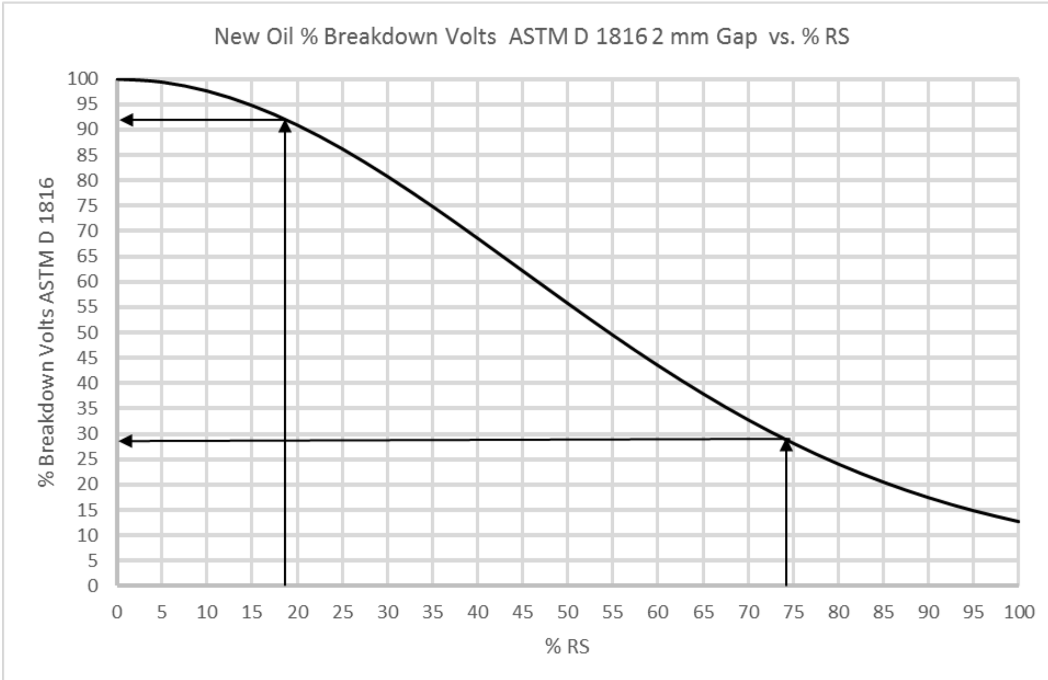

Figure B.1—Normalized voltage breakdown vs. %RS

Example: As an example of the application of these references and the graph in Figure B.1, consider mineral oil with a temperature of 30 °C and a dissolved water level of 15 ppm at equilibrium. Application of Equation (B.1) with Oommen's parameters for solubility gives 18% for %RS. This gives a normalized breakdown voltage of 92 % in Figure B.1.

However, at 0 °C, the same equation gives a 74%RS. This gives a normalized breakdown voltage of 28%. In terms of reenergizing a transformer at 0 °C, the dielectric breakdown voltage, in this example, could be reduced to 32% of the original value (i.e., 28%/92% = 0.32 multiplier).

In this example, an ASTM D1816 (2 mm gap) voltage breakdown value of 50 kV for 15 ppm oil at 30 °C is not the same as it would be at 0 °C (both at temperature equilibrium conditions). The voltage breakdown value could be reduced down to 15 kV (50 kV × 0.32 = 15 kV). In this example, there is a concern about mineral oil dielectric breakdown failure at 0 °C. In this case, IEEE Std C57.93 should be consulted to provide more information on suggested procedures for energization of mineral oil transformers under cold conditions.

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